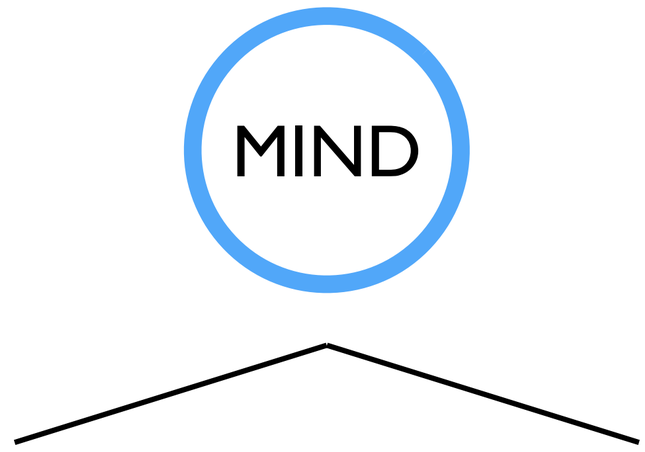
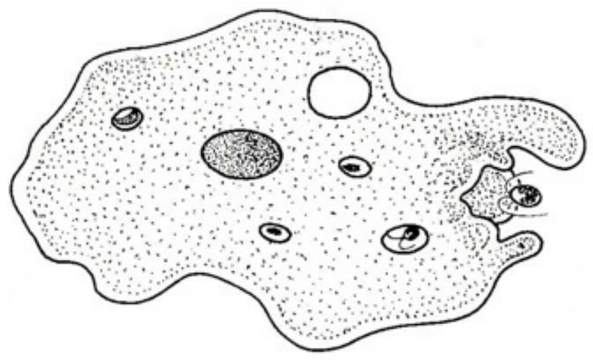
35. Physical Cognition

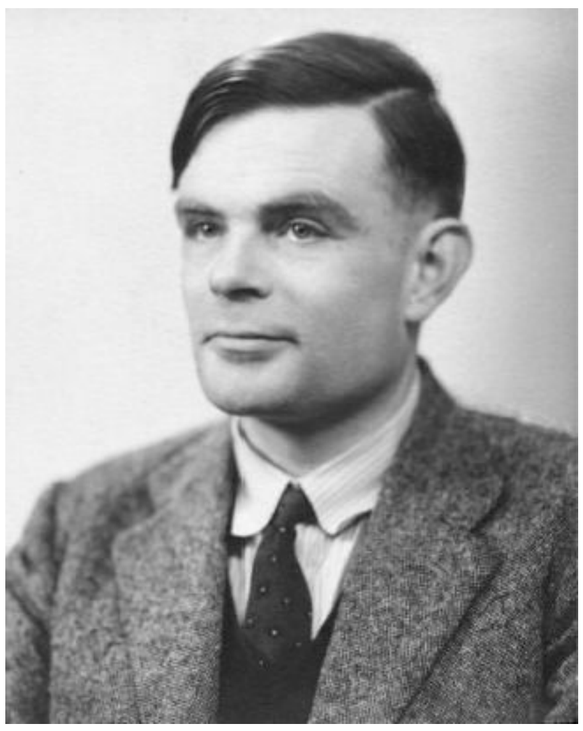
"A belief … is a map of neighbouring space by which we steer."
—Ramsey 1931









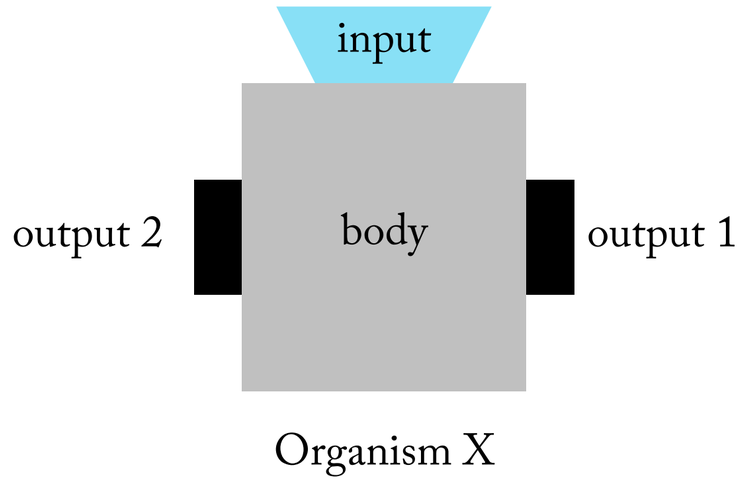


From Object to Animal

Suppose we found a strange object in the woods and attempted to understand its behavior.



After much experimentation, we determine that it is some sort of input-output system. Perhaps it is even an animal of some unknown kind. We'll call it Organism X.



Organism X

To study it's behavior, we schematize it's overall organization into four basic components: an input organ, two output organs, and a body connecting them.

Does Organism X have cognition? Does it think? Have beliefs? Consciousness? To put it to the test, we stimulate the input organ with a series of signals over time and record the output. Here are some snapshots of one of our experiments.

…	t13	t12	t11	t10	t9	t8	t7	t6	t5	t4	t3	t2	t1	
…	0	0	0	0	0	1	0	0	0	0	0	0	0	IN
…	11	11	11	01	11	01	11	11	11	11	11	11	10	OUT

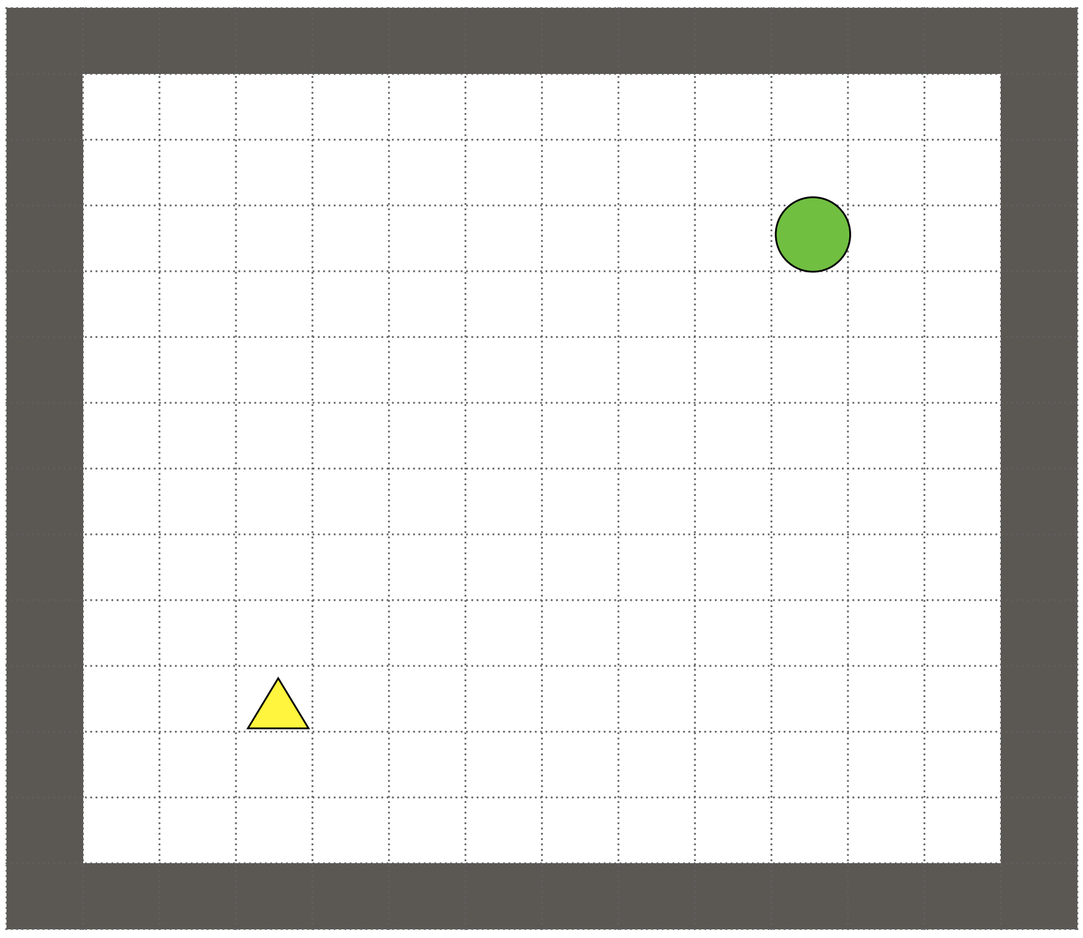
…	t111	t110	t109	t108	t107	t106	t105	t104	t103	t102	t101	t100	…	
…	0	0	0	0	0	0	*	0	0	1	0	0	…	IN
…	11	11	11	11	11	01	11	11	11	10	11	11	…	OUT

…	t198	t197	t196	t195	t194	t193	t192	t191	t190	t189	t188	t187	…	
…	0	0	0	0	0	0	0	0	0	0	0	0	…	IN
…	00	00	00	00	00	00	00	11	11	11	11	11	…	OUT

There is certainly something complex about Organism X's reactions. But are they random or rational? Is it trying to tell us something? Do the inputs *mean anything* to it?

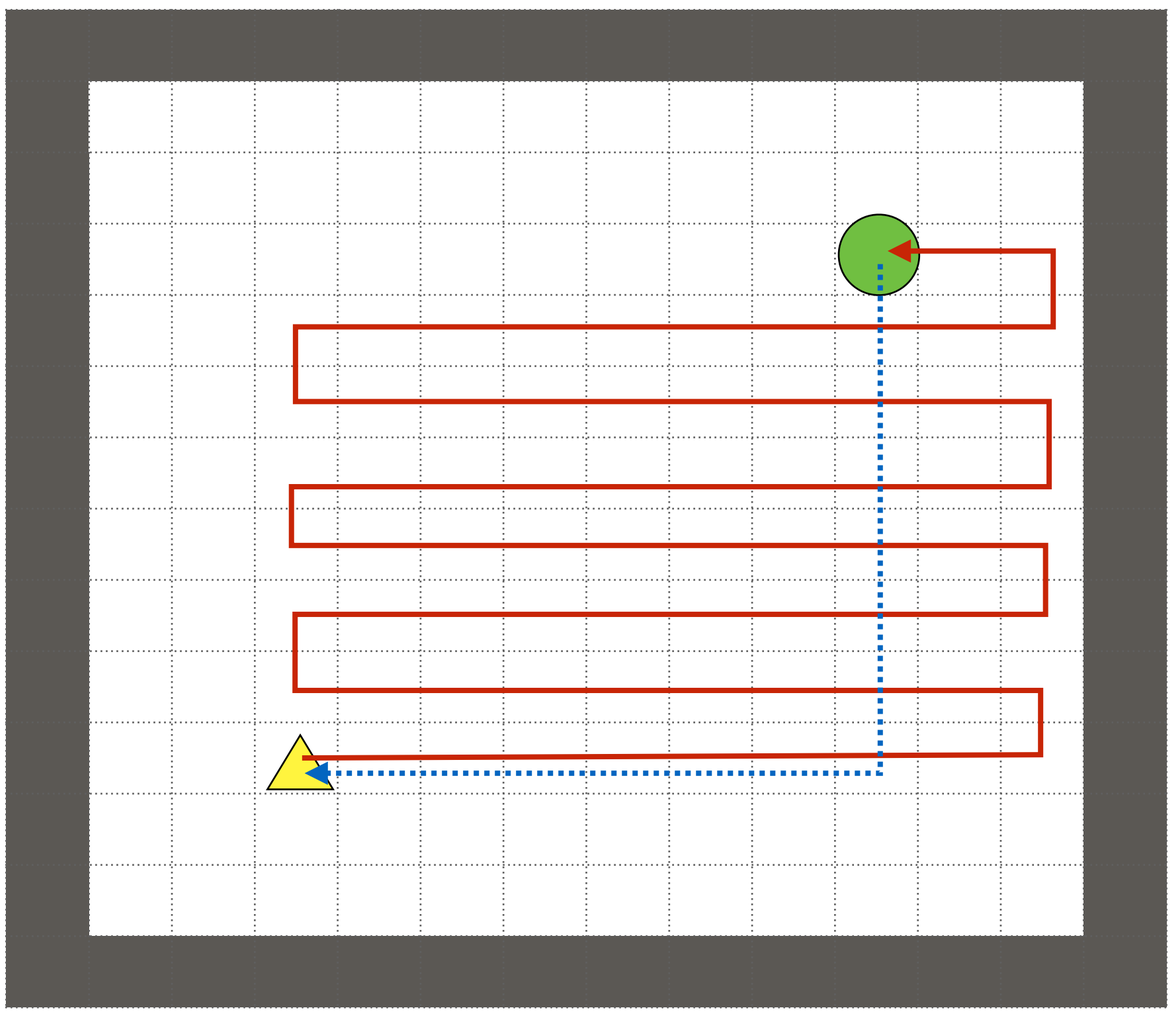
The Ecological View

We consult with a cognitive scientist, who gives us some wise advice: you won't find any answers to your questions unless you look at the **ecological environment** in which Organism X evolved. Only then can we attempt to decode its inputs and outputs.



Consulting with some naturalists, it turns out that X's tend to live in enclosed spaces, where they have single food source— a kind of plant that lives only a single day— and no predators. In other words, the X ecology looks a bit like this.

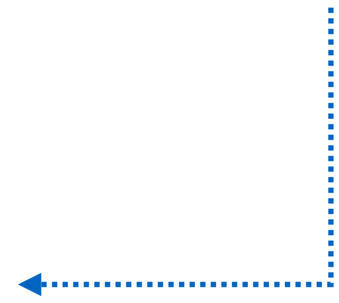
Observed Behavior





Phase 1

Phase 2



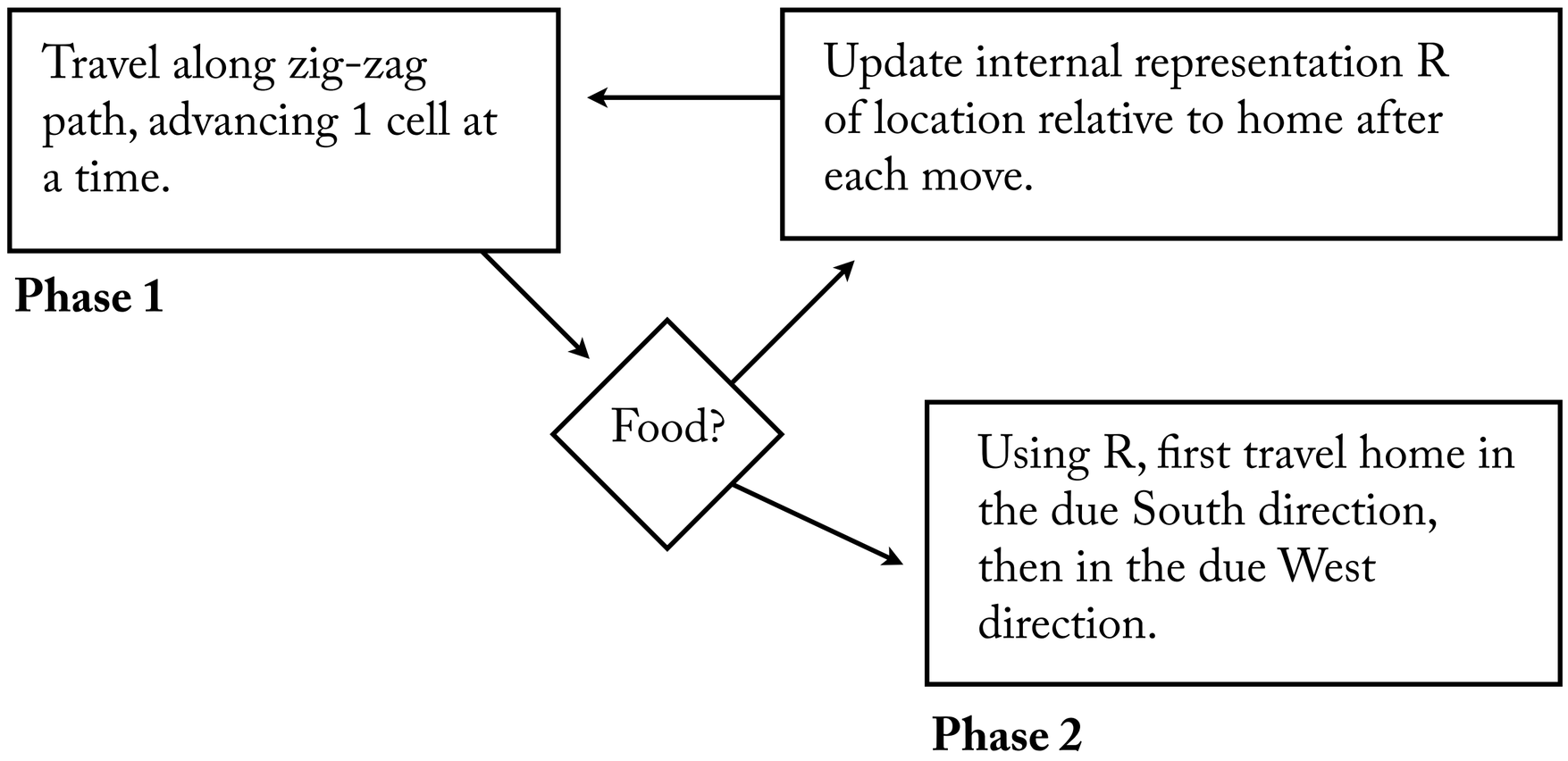
Evolutionary Analysis

Phase 1:
Exhaustive search for food through NE quadrant.

→

Phase 2:
Return home as efficiently as possible.

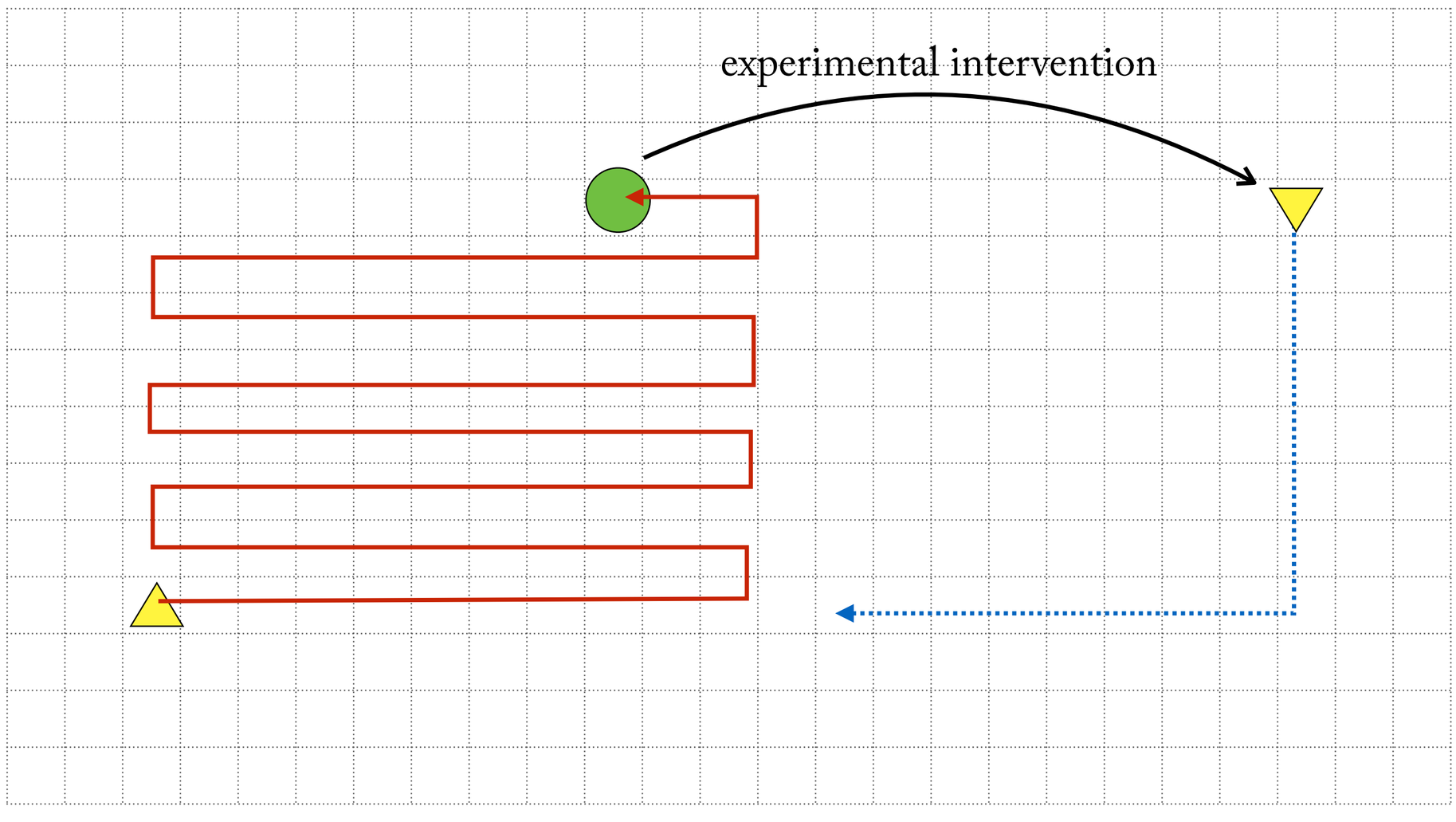
The Representational Hypothesis



"What makes this of interest to cognitive scientists is that it suggests that the brain of an insect is a symbol processing organ. It suggests that the brain uses computation to construct a cognitive map and to compute courses from the positional values stored in that map. The positional data stored on a map together with the computations that utilize those data to set courses constitute a **representation**, a model of the world used to determine courses of action in that world. Thus, the homing abilities of insect might be taken as strong evidence in favor of a computational-representational theory of mind."

Testing the Representational Hypothesis

An experiment with Turbots: evidence for mental representations.



Neuroscience

Next we capture a Turbot, bring it to the lab, and look inside. When we inspect the tape, this is what we find:

*	0	0	0	0	1	*	0	0	0	1	0	*

We capture other Turbots, find variations of our first result.

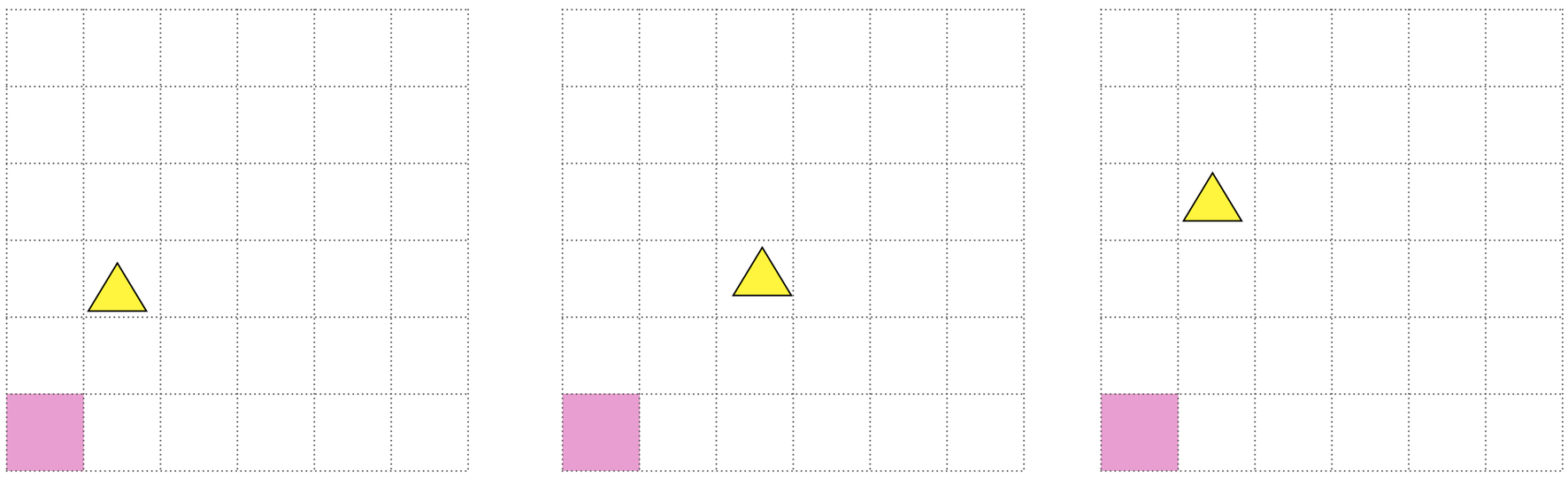
*	0	0	0	0	1	*	0	0	0	1	0	*

*	0	0	0	1	0	*	0	0	0	1	1	*

There are clearly some stable patterns here, but also variation. What does this all mean?

Cognitive Science

After testing several different hypotheses, someone in our lab comes up with the idea that the mark on the tape are correlated to the X- and Y- position of the Turbot. This makes sense from an evolutionary perspective. So we capture three different Turbots at different stages of their search for food.



00001 00010 00010 00010 00001 00011

We still don't really know what these marks *mean*, but we have a clue. When the Turbot moves in the X- direction, the left-hand block changes; when the Turbot moves in the Y-direction, the right-hand block changes. Otherwise they stay the same. This gives rise to the following hypothesis:

*	0	0	0	0	1	*	0	0	0	1	0	*

records X-position records Y-position

Externalism

According to **externalism,** for an internal symbol S to be a genuine mental representation of X, it must satisfy two criteria:

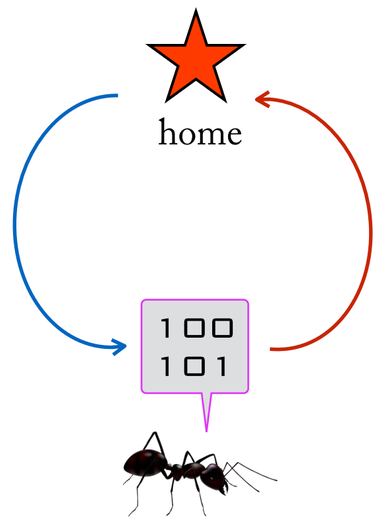
(1) it must function to mirror X in the world;

(2) it must be guided by perception (of X) and guide action (towards X).

When these criteria are met, mental representations are capable of **truth** and **falsity.**

Navigational symbols have meaning for the ant because:

- internal symbols c**arry information** about the direction home.
- internal symbols **guide the action** of the ant with respect to the direction home



Information Semantics

Mirroring

Further study reveals that the symbols in the Turbot memory systematically covary with the physical distance between the turbot and home. In fact, when the system is working properly, the following generalizations hold:

> binary(S_x) = the number of steps **east** between **here** and **home**.
> binary(S_y) = the number of steps **north** between **here** and **home**.

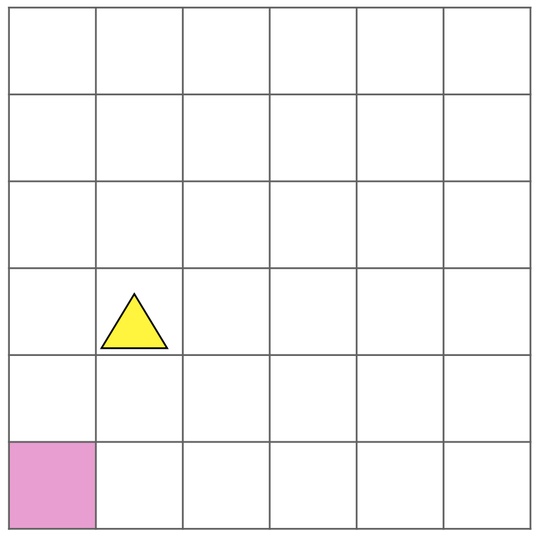
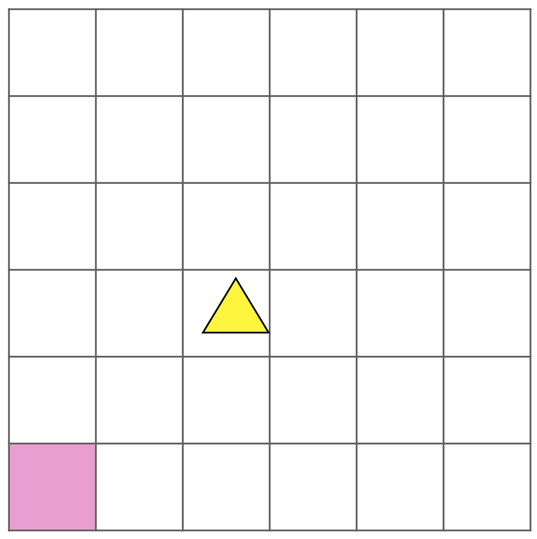
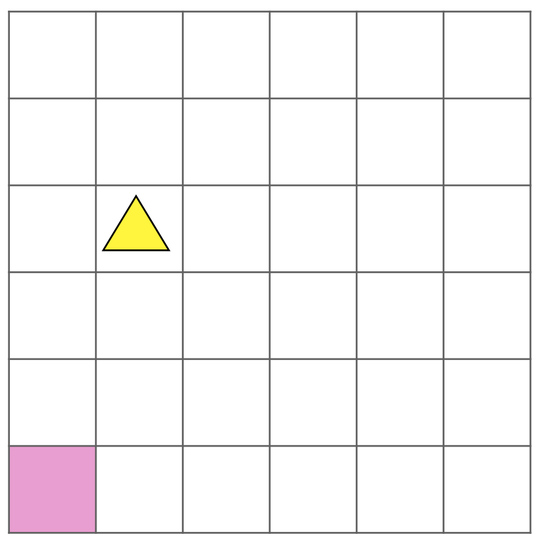
Mediating action and perception.

Truth and Falsity

This leads the idea that these symbols are **mental representations:** internal symbols which represent the "beliefs" or "cognitive commitments" of the Turbot. At any given time, the Turbot has the "beliefs":

> The number of steps **east** between **here** and **home** is binary(S_x).
> The number of steps **north** between **here** and **home** is binary(S_y).

These beliefs may be **true** or **false**. As we saw in our original displacement experiment, it is clear that the Turbots follow their internal symbols, *as if* the equations above were true, even when they are not.

Internal world	External world
00001 00010	
00010 00010	
00001 00011	

"A *mental representation* is a functioning isomorphism between a set of processes in the brain and a behaviorally important aspect of the world…"

Mental Representation

Further study reveals that the symbols in the Turbot memory systematically covary with the physical distance between the turbot and home. In fact, when the system is working properly, the following generalizations hold:

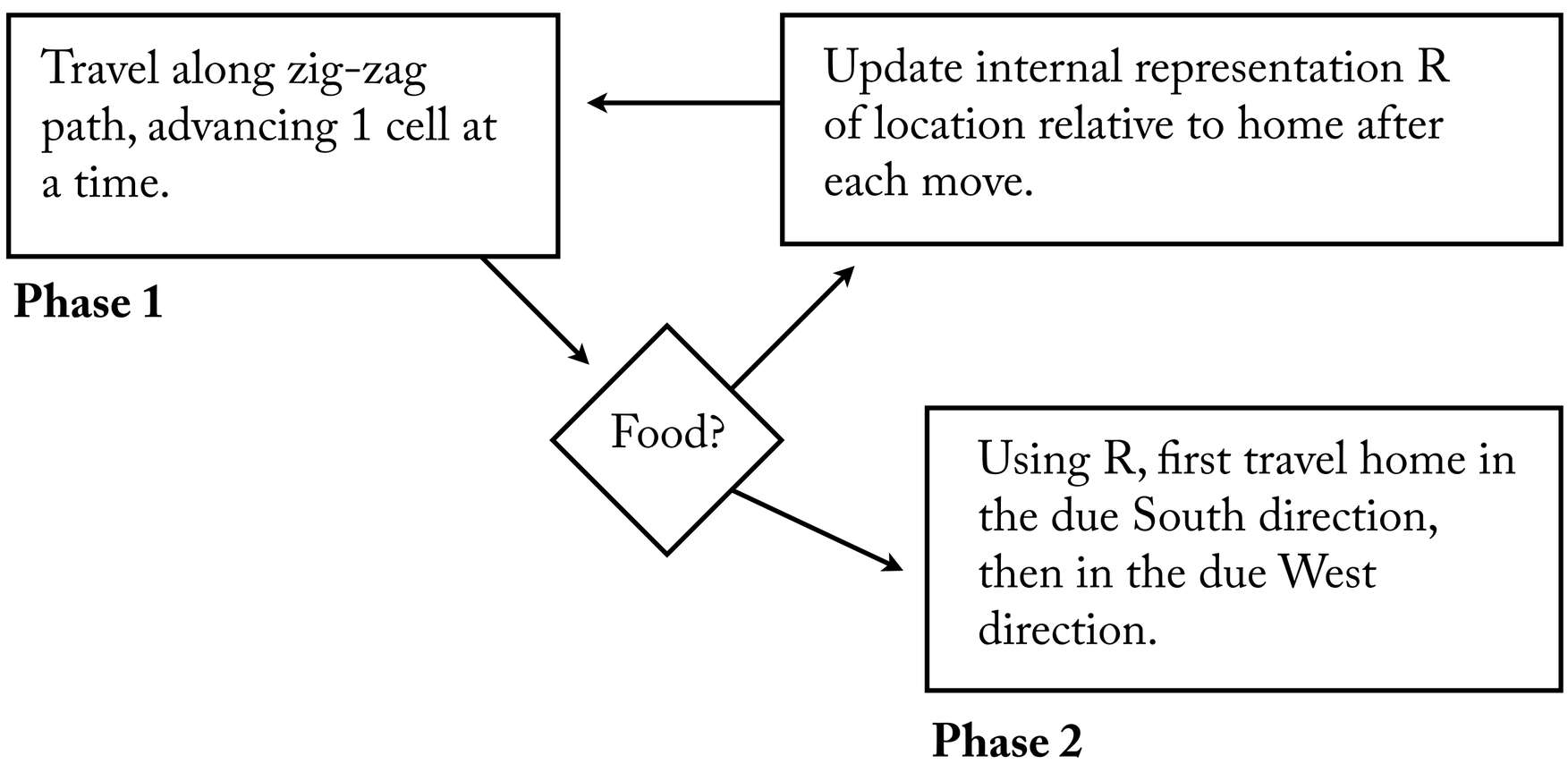
$\text{distance}_x(\text{here, home}) = \text{Bin}(S_x)$
$\text{distance}_y(\text{here, home}) = \text{Bin}(S_y)$

Yet, as we saw in our original displacement experiment, the system is not always working properly. Still, it is clear that the Turbots follow their internal symbols, *as if* the equations above were true, even when they are not.

This leads the idea that these symbols are **mental representations:** internal symbols which represent the "beliefs" or "cognitive commitments" of the Turbot.

Representational content = (at a time)
$\text{distance}_x(\text{here, home}) = a$ &
$\text{distance}_y(\text{here, home}) = b$

where $a = \text{Bin}(S_x)$ and $b = \text{Bin}(S_y)$



Mirroring the world. These representations can be thought of creating an internal way of "tracking" or "mirroring" an external phenomena.

Truth and falsity. The machine misrepresents the environment when the content of its symbols is false.

"A *mental representation* is a functioning isomorphism between a set of processes in the brain and a behaviorally important aspect of the world…

"Is the brain a symbol-processing system? The rise of cognitive psychology in the sixties reflected the emergence of the (then novel) view that the brain was a symbol-processing system analogous to a computer, and that it used its symbol-processing capacity to construct representations of the environment in which it functioned."

Suppose that we discover some Turbots in the wild, and we attempt to describe them as a cognitive scientist would.

Suppose further that they are the products of natural selection: over millions of cycles of reproduction, mutation, and selective pressure, our turbots have evolved. We assume that:

- **The environment** has stayed the same over the course of natural selection (it always involves the surrounding walls, and the placement of food).
- **Reproduction** involves the duplication of machines in similar environments.
- **Variation** between machines involves random permutations of logic gates or tape configurations.
- **Selective pressure**: if a turbot doesn't find food and return home within a certain time span, it dies without having a chance to reproduce.